Solar System Models · Knowledge Organiser

Learning goal: I can use solar system data and a scale model to link distance from the Sun to orbit time.

Today I Can

I can read planet distance and orbit time from a solar system data table.

I can describe the pattern linking distance from the Sun to the time for one orbit.

I can state one strength and one limitation of a classroom solar system model.

Must-Know Words

orbit time: the time a planet takes to travel once all the way around the Sun (its year)

scale model: a model that shrinks huge real distances or sizes using a fixed rule

astronomical data: measurements of planets such as distance from the Sun, diameter, and orbit time

limitation: something a science model leaves out or cannot show accurately

Core Science Steps

First, look at a 1-metre paper strip model: the four rocky planets cluster in the first 5 cm near the Sun!

Second, the four giant planets spread out across the rest of the strip all the way to 100 cm (Neptune).

Third, check the data table: every step further from

the Sun gives a longer orbit time.

Keep the Science Clear

Use the data table to quote two planets' distances and orbit times, then evaluate what the strip model shows and misses.

A paper strip shows scaled distance from the Sun, but it cannot show planet diameters to the same scale or moving 3D orbits.